Home Office Composition, Fit & Budget Logic

AID-LOG-005 | Revision 1

Purpose. Defines the backend logic used by AIDOOi to translate Home Office inputs into an appropriate workspace composition, validate product fit and circulation, apply room-level budget logic, and generate recommendations. Customer journey inputs are defined in AID-SPC-005.

> **Core sequence: Room Dimensions -> Room Size -> Number of Workspace Users -> Desk / Workspace Fit -> Office Chair Fit & Circulation -> Remaining Composition -> Budget -> Recommendation.**

1. Core Principles

1. Desk / Workspace and Office Chair form the Anchor Workspace. The number of required workspaces is driven by the customer's selected Number of Workspace Users.

2. Required Workspace Capacity has priority over decorative completeness. AIDOOi must not silently reduce the number of users to make secondary furniture fit.

3. Desk Configuration may use individual desks or an appropriate shared workstation depending on user count and available placement space.

4. Chair movement and circulation must be validated after desk placement. Geometric desk fit alone is insufficient.

5. Default Composition is a recommendation baseline, not a hard constraint. Secondary furniture may be reduced or removed when the layout does not support it.

6. In Compact Home Offices, vertical storage should be preferred over unnecessary use of floor space.

7. Larger offices should create better functional zoning rather than simply contain more furniture.

8. Style and Colour are matching inputs under AID-DATA-009 and AID-DATA-007. They do not directly define the numeric budget range.

9. Room Budget Tier is not the same as Product Price Tier. Product tiers may be mixed to preserve workspace capacity, fit and the overall room budget.

10. Final Composition Price equals the sum of Current Selling Prices of selected products and recalculates dynamically when products are removed.

2. Home Office Size Classification

Room Area	Classification
< 6 m²	Compact
6-10 m²	Small
10-15 m²	Medium
15-22 m²	Large
22+ m²	Extra Large

Room Area is system-derived from the Length and Width supplied in AID-SPC-005. The customer does not select this classification.

3. Fit Priority & Product Dependencies

> **Base fit priority: Required Workspaces -> Office Chairs -> Chair Movement / Circulation -> Storage -> Task Lighting -> Secondary Functional Zone -> Decor.**

Dependency	Logic
Workspace Quantity	Derived from customer-selected Number of Workspace Users.
Desk Configuration	Derived from Number of Users plus available wall / placement space. Individual desks and shared workstations are both valid.
Office Chair Quantity	One appropriate office chair per required workspace user.
Task Lighting	Scaled to the number and configuration of workspaces.
Storage	Added after anchor workspace fit and circulation are secured; vertical solutions are preferred in compact rooms.
Secondary Seating	Conditional. Added only after complete workspace requirements, storage and circulation are satisfied.

Workspace Constraint Rule. If the selected number of users cannot be accommodated with acceptable workspace fit and circulation, AIDOOi must surface a Fit Constraint rather than automatically reducing the number of users.

4. Composition by Room Size

4.1 Compact Home Office (< 6 m²)

Product	Qty	Preferred Fit Logic
Desk	1	Compact, approximately 80-120 cm wide and 45-60 cm deep.
Office Chair	1	Compact; sufficient space required for chair movement.
Task / Desk Lamp	1	Compact; wall-mounted may be used where desk surface is limited.
Bookcase / Shelving	0-1	Vertical / compact preferred; conditional on circulation.
Storage Cabinet / Drawers	0-1	Compact; only where placement does not obstruct movement.
Wall Art / Decor	1-2	Primarily wall-based to preserve floor space.

Fit Priority: Desk -> Office Chair -> Chair Movement / Circulation -> Storage -> Task Lighting -> Decor

Compact-space rule: Prefer vertical storage over floor-consuming storage wherever practical.

4.2 Small Home Office (6-10 m²)

Product	Qty	Preferred Fit Logic
Desk / Workspace	1-2	Approximately 100-140 cm wide per workspace, or an appropriate shared desk.
Office Chair	Selected users	One per workspace user.
Task / Desk Lamp	1-2	Aligned to workspace configuration.
Bookcase / Shelving	1	Approximately 60-100 cm wide; vertical storage preferred.
Storage Cabinet / Drawers	1	Compact / Medium; must preserve chair movement.
Wall Art / Decor	1-2	Primarily wall-based.
Planter / Floor Decor	0-1	Only if an appropriate floor zone remains.

Fit Priority: Required Workspaces -> Office Chairs -> Chair Movement / Circulation -> Storage -> Lighting -> Decor

4.3 Medium Home Office (10-15 m²)

Product	Qty	Preferred Fit Logic
Desk / Workspace	1-2	Approximately 120-160 cm wide per workspace; individual or shared.
Office Chair	Selected users	One per workspace user.
Task / Desk Lamp	1-2	Aligned to workspace configuration.
Bookcase / Shelving	1	Approximately 80-120 cm wide.
Storage Cabinet / Drawers	1	Medium; based on wall / placement zone.
Accent Chair	0-1	Conditional; only after workspace and circulation are satisfied.
Side Table	0-1	Conditional with secondary seating.
Wall Art / Decor	2-3	Based on wall and surface zones.
Planter	0-1	Only where an appropriate floor zone exists.

Fit Priority: Required Workspaces -> Office Chairs -> Chair Movement / Circulation -> Storage -> Lighting -> Secondary Seating -> Decor

Secondary Seating Rule: Secondary seating is conditional, not mandatory.

4.4 Large Home Office (15-22 m²)

Product	Qty	Preferred Fit Logic
Desk / Workspace	1-3	Approximately 140-180 cm for primary workspace; configuration based on users.
Office Chair	Selected users	One per workspace user.
Task / Desk Lamp	By workspaces	Aligned to desk configuration.
Bookcase / Shelving	1-2	Approximately 80-140 cm per unit.
Storage Cabinet / Drawers	1-2	Based on wall zones and composition need.
Accent Chair	1-2	For a conditional secondary seating zone.
Side Table	0-1	Associated with secondary seating.
Floor Lamp	0-1	Conditional for reading / seating zone.
Wall Art / Decor	3-4	Based on wall and surface zones.
Planter	1	Appropriate floor / corner zone required.

Fit Priority: Required Workspaces -> Office Chairs -> Chair Movement / Circulation -> Storage -> Lighting -> Secondary Seating -> Decor

Functional Zoning Rule: A larger office should create better functional zones, not simply contain more furniture.

4.5 Extra Large Home Office (22+ m²)

Product	Qty	Preferred Fit Logic
Desk / Workspace	1-3+	Approximately 140-200 cm for primary workspace; configuration based on users.
Office Chair	Selected users	One per workspace user.
Task / Desk Lamp	By workspaces	Aligned to desk configuration.
Bookcase / Shelving	1-2	Approximately 100-160 cm per unit.
Storage Cabinet / Drawers	1-2	Based on storage and wall zones.
Accent Chairs	2	For conditional meeting / reading zone.
Side Table	1	For secondary seating zone.
Floor Lamp	1	Conditional for reading / seating zone.
Rug	0-1	May define a secondary zone; not mandatory.
Wall Art / Decor	4-6	Based on wall and surface zones.
Planter	1-2	Only in suitable floor / corner zones.

Fit Priority: Required Workspaces -> Office Chairs -> Chair Movement / Circulation -> Storage -> Task Lighting -> Secondary Zone -> Decor

Secondary Functional Zone Rule: A meeting / reading zone may be created only after the complete workspace requirement has been satisfied. Extra room area does not automatically justify extra furniture.

5. Home Office Budget Logic

Budget is applied at room-composition level. Room Size defines the expected composition and fit envelope, while Number of Workspace Users acts as an important cost modifier because it affects desk, office chair and task-lighting requirements.

> **Budget relationship: Room Size + Number of Workspace Users -> Composition -> Budget Tier -> Style & Colour Matching -> Recommendation.**

The customer-facing budget tiers are Essential, Value, Standard, Premium and Luxury. Numeric ranges below are target ranges for the complete recommended Home Office composition.

Room Size	Essential	Value	Standard	Premium	Luxury
Compact (<6 m²)	500-1,200	1,200-2,000	2,000-3,500	3,500-6,000	6,000+
Small (6-10 m²)	800-1,800	1,800-3,000	3,000-5,000	5,000-8,000	8,000+
Medium (10-15 m²)	1,200-2,500	2,500-4,000	4,000-6,500	6,500-10,500	10,500+
Large (15-22 m²)	2,000-3,500	3,500-5,500	5,500-9,000	9,000-15,000	15,000+
Extra Large (22+ m²)	3,000-5,000	5,000-8,000	8,000-13,000	13,000-22,000	22,000+

User-count Modifier. Two offices in the same size class can produce different actual composition prices when their required workspace-user counts differ. The user count must be reflected in desk configuration, office chair quantity, task lighting and budget allocation.

> **Budget Allocation Priority: Required Workspaces -> Office Chairs -> Storage -> Task Lighting -> Secondary Functional Zone -> Decor.**

If required workspaces consume a larger share of the selected budget, AIDOOi may use lower Product Price Tiers for secondary items. It must not reduce the required workspace count or compromise fit merely to preserve secondary furniture.

Budget Tier Principle: A Budget Tier defines a target spending range, not a spending requirement. AIDOOi must not add unnecessary furniture or choose more expensive products solely to reach the lower boundary of a selected tier.

Product-level prices and tier logic reference AID-LOG-001. Final room price uses Current Selling Price.

6. Recommendation Rules

1. Validate all required workspaces and office-chair movement before selecting secondary furniture.
2. Respect the customer's selected Number of Workspace Users unless a Fit Constraint must be surfaced.
3. Choose between individual desks and shared workstations according to user count and available placement space.
4. Use Style and Colour preferences to rank products that already satisfy fit, availability and budget logic.
5. Allow product-tier mixing inside a room budget tier.
6. Use Current Selling Price when calculating the final composition total.
7. Recalculate the total dynamically when the customer removes products.
8. Do not overfill larger offices solely to consume space or budget.

7. Standard References

Reference	Standard
AID-SPC-005	Home Office Path Standard
AID-DATA-007	Product Colour Standard
AID-DATA-009	Product Style Standard
AID-LOG-001	Product Price Range Standard
AID-CAT-001	Seating Standard
AID-CAT-003	Tables Standard

Reference	Standard
AID-CAT-005	Storage & Casegoods Standard
AID-CAT-006	Lighting Standard
AID-CAT-007	Rugs & Soft Furnishings Standard
AID-CAT-008	Decor & Accessories Standard

Revision 1 status: Approved MVP logic. Numeric thresholds and composition preferences may be revised later using observed inventory, user behavior and recommendation performance.